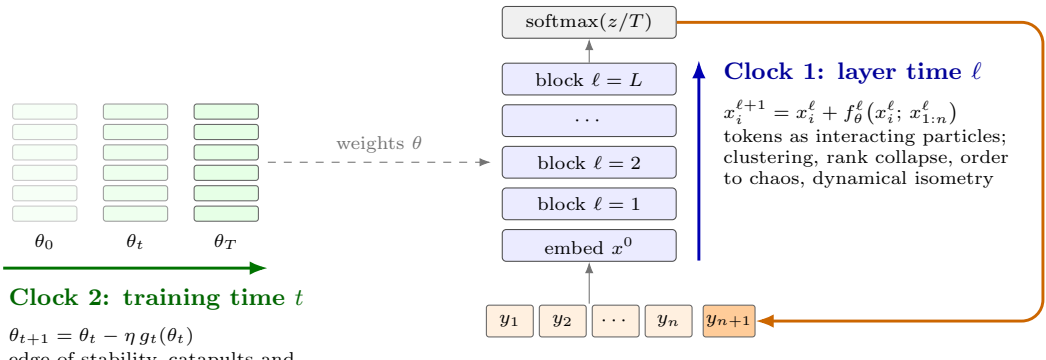


Clock 1: layer time ℓ

$x_i^{\ell+1} = x_i^\ell + f_\theta^\ell(x_i^\ell;\, x_{1:n}^\ell)$
tokens as interacting particles; clustering, rank collapse, order to chaos, dynamical isometry

Clock 2: training time t

$\theta_{t+1} = \theta_t - \eta\, g_t(\theta_t)$
edge of stability, catapults and loss spikes, saddle plateaus, grokking

Clock 3: generation time n

$y_{n+1} \sim \mathrm{softmax}\big(z(y_{1:n};\theta)/T\big)$: repetition loops as attractors, temperature transitions, drift, collapse